GLOBAL WILDFIRE MONITORING & FORECASTING PORTAL

Simple Explanation of the Complete System

This document explains the wildfire portal from the beginning in simple language: what the system does, what data it needs, how the data moves through the system, and how monitoring becomes forecasting.

Main idea: First find fires → understand the environment → learn from historical fires → predict new fires → predict existing-fire spread → predict smoke → show everything on one map.

Phase 1 — Build the Foundation

Goal: Create the technical base before advanced forecasting.

What to build

Backend: Python + FastAPI. Database: PostgreSQL + PostGIS. Large-file storage: S3-compatible object storage. Frontend: React + MapLibre. Scheduler: Airflow, Prefect or Dagster.

Why PostGIS?

Your system needs geographic questions such as: find fires within 50 km, identify land cover at a fire, or find nearby roads.

Storage

Keep large satellite/weather/raster files in object storage. Keep geographic records, metadata and indexes in PostgreSQL/PostGIS.

Result

You have a working backend, database, file storage, map and scheduled-job framework.

Phase 2 — Live Global Fire Monitoring

Goal: Show where fires are being detected now.

Starting source

Use NASA FIRMS / VIIRS as the first active-fire source: https://firms.modaps.eosdis.nasa.gov/

Important parameters

Latitude, longitude, detection time, satellite/sensor, confidence, Fire Radiative Power (FRP), day/night information and quality flags.

Detection vs fire event

A satellite may detect many nearby points belonging to one wildfire. Keep every original detection, but group related detections into a higher-level fire event.

Fire-event information

Store a permanent fire_event_id, center location, first/latest detection, number of detections, maximum FRP and history.

Result

The portal can show global fires, time filters and a details panel for each event.

Phase 3 — Land Cover and Terrain

Goal: Understand what physically surrounds each fire.

Land cover

Use a Copernicus global land-cover product such as the LCM-10 baseline. Treat the 2020 baseline as a reference layer, not complete truth about 2026 vegetation condition: https://land.copernicus.eu/

What it tells you

It identifies forest, grassland, shrubland, cropland, wetland, urban land, bare land and other classes.

Terrain

Use a DEM such as Copernicus DEM. Calculate elevation, slope and aspect; later add curvature/ruggedness: https://dataspace.copernicus.eu/

Important distinction

Land cover tells you the surface type. It does not tell you exactly how dry the vegetation is today. Current vegetation and moisture must be added separately.

Result

Every fire can be enriched with land-cover class, elevation, slope and aspect.

Phase 4 — Weather, Vegetation and Moisture

Goal: Understand current and future environmental conditions.

Historical weather

Use ERA5 for historical analysis and training: https://cds.climate.copernicus.eu/

Weather parameters

Temperature, relative humidity, dew point, wind speed, wind direction, wind gust, precipitation, pressure and solar radiation.

Future weather

Use a forecast source such as ECMWF Open Data or another suitable provider. Keep forecast issue time separate from valid time.

Vegetation

Use Sentinel-2 imagery to calculate NDVI, NDMI and NBR. Filter clouds and poor-quality observations: https://documentation.dataspace.copernicus.eu/APIs.html

Soil moisture

Use products such as NASA-USDA Global Soil Moisture; drought indices can be added later: https://earth.gsfc.nasa.gov/hydro/data/nasa-usda-global-soil-moisture-data

Result

The portal can explain whether an area is hot, dry, windy, recently wet or vegetation-stressed.

Phase 5 — Build the Historical Fire Database

Goal: Create the examples needed to train and evaluate forecasting models.

Historical information

Collect historical fire detections, perimeters/burned areas where available, weather, vegetation, moisture, land cover, terrain, lightning and later human-activity information.

Common grid

Divide the study area into regular cells such as 1 km, 2 km or 5 km depending on the use case and data resolution.

Prediction label

For each cell and prediction time, mark whether a new fire occurred during the future window, such as the next 24 hours.

Most important ML rule

Never give the model information that would not have been available at prediction time. Future observations are labels, not inputs. This prevents data leakage.

Result

You have a versioned historical dataset that can be regenerated from documented source data.

Phase 6 — Fire-Risk Forecasting

Goal: Predict where a new fire is more likely to occur in the next 24, 48 or 72 hours.

Inputs

Weather: temperature, humidity, wind, gusts, rainfall and dryness indicators. Vegetation: NDVI, NDMI and anomalies. Moisture: soil moisture/drought. Also land cover, terrain, lightning, fire history and human activity.

Start simple

Begin with a baseline model, then test LightGBM or XGBoost for structured environmental data.

Output

For each grid cell, produce a probability or calibrated risk value. The interface can later translate values into understandable categories using validated thresholds.

Forecast flow

Current data + future weather → features → ML model → probability per grid cell → risk raster → map/API → dashboard.

Evaluation

Use time-based and geographic validation. Useful metrics include precision-recall AUC, recall, Brier score and calibration.

Result

The portal automatically produces versioned 24/48/72-hour risk maps and compares forecasts with later observations.

Phase 7 — Fire-Spread Forecasting

Goal: Estimate where an existing wildfire may move.

Different problem

Phase 6 asks where a new fire might start. Phase 7 asks where an existing fire might move.

Inputs

Current perimeter or hotspot cluster, recent growth, future wind, temperature, humidity, precipitation, fuel/land cover, vegetation/moisture and terrain.

Build in stages

Start with a simple spread prototype. Run 6h, 12h, 24h, 48h and 72h scenarios. Compare predictions with later observed boundaries and improve using historical fires.

Uncertainty

Fire spread is uncertain. Show a forecast boundary together with uncertainty or scenarios rather than treating it as exact.

Result

A selected fire can display its current perimeter and possible future spread areas.

Phase 8 — Smoke, Alerts and Production

Goal: Turn the system into an operational portal.

Smoke

Integrate an atmospheric composition/forecast source such as CAMS: https://www.ecmwf.int/en/forecasts/datasets/cams-global-atmospheric-composition-forecasts

Smoke workflow

Combine fire location/intensity with atmospheric conditions and display smoke or air-quality information where appropriate.

Alerts

Let users monitor an area. Generate alerts for new fires, validated high-risk conditions, or an existing fire approaching a monitored/populated area.

Production

Add backups, source-failure monitoring, API rate limits, model monitoring, audit logs, performance optimization, load testing and recovery documentation.

Result

The portal can operate continuously and provide monitoring, forecasts, smoke information and alerts.

How One Fire Moves Through the System

Step	What happens
1	Satellite detects a fire location.
2	FIRMS connector downloads the observation.
3	Original data is saved for provenance.
4	The system validates coordinates, time, confidence and other fields.
5	The detection is stored in PostGIS.
6	Nearby detections are grouped into a fire event.
7	Spatial processing finds land cover, terrain, weather, vegetation, moisture and other context.
8	The enriched fire appears in the portal.
9	Forecasting models use the appropriate current and future information.
10	Risk, spread and smoke results are published through APIs and map layers.

The Main Data Flow

External sources → data connectors → validation/processing → database + object storage → environmental enrichment → forecasting → APIs/map layers → web portal.

Architecture rule: Do not make the frontend directly depend on every external provider. Put provider-specific download/API logic inside separate connector modules, then expose a consistent internal API to the frontend.

Conceptual Database

Category	Purpose
Fire detections	Original satellite observations.
Fire events	Grouped wildfire incidents.
Fire perimeters	Observed fire boundaries where available.
Land cover	Surface type and product version.
Weather observations	Historical atmospheric conditions.
Weather forecasts	Future atmospheric conditions with issue/valid times.
Vegetation	NDVI/NDMI/NBR and related indicators.
Soil moisture	Ground moisture and dryness.
Terrain	Elevation, slope and aspect.
Lightning	Potential ignition-related observations.
Human activity	Roads, settlements, agriculture and similar context.
Smoke	Smoke/aerosol forecasts and observations.
Forecast results	Risk and spread model outputs.
Alerts	User monitoring and notification records.

Recommended Development Order

1. Database + object storage + ingestion framework
2. VIIRS/FIRMS live monitoring
3. Fire-event clustering
4. Global map and fire details
5. Land cover + DEM
6. ERA5 + future weather
7. Sentinel-2 + soil moisture
8. Historical training dataset
9. 24/48/72-hour fire-risk model
10. Fire-spread model
11. Smoke/CAMS
12. Alerts + production hardening

The Project in One Sentence

Build a system that continuously collects satellite fire observations and environmental data, combines them into a common geographic database, uses historical data and weather forecasts to estimate future fire risk and spread, and presents the results through a global interactive map.

The Eight Phases at a Glance

Phase	Purpose
1	Build the technical foundation.
2	Show fires happening now.
3	Understand land and terrain.
4	Add weather, vegetation and dryness.
5	Build historical fire data.
6	Predict where new fires may start.
7	Predict how existing fires may spread.
8	Add smoke, alerts and production features.

Final Developer Rule

Do not start with AI. First make the data pipeline reliable. Every fire should have a source, time, location and quality information. Then enrich it with environmental information. Then create the historical training dataset. Only after that should the forecasting model be trained.